

## PAMPA, TX, USA

WMO: 720313

Lat: 35.613N

Lon: 100.996W

Elev: 989

StdP: 89.99

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03052

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.3      | -9.5 | -20.0                         | 0.7 | -9.0 | -16.7 | 1.0 | -3.4 | 14.7                  | 9.3  | 13.5 | 10.0 | 4.5                   | 10   | 0.650 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.7                   | 37.3            | 19.0 | 35.9 | 19.3 | 34.5 | 19.3 | 22.6                | 30.5 | 21.8 | 29.8 | 21.2 | 29.1 | 6.2                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 20.6                            | 17.2 | 25.3 | 19.9 | 16.5 | 24.7 | 19.0 | 15.6 | 24.2 | 72.2          | 30.8 | 69.0 | 29.9 | 66.4 | 28.8 | 26.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 13.7              | 12.0 | 10.9 | -17.0                      | 39.8 | 2.3                | 1.9 | -18.7                                              | 41.2 | -20.0      | 42.3 | -21.3      | 43.3 | -23.0      | 44.7 |    |    |
|                   |      |      | -17.5                      | 23.3 | 2.2                | 1.2 | -19.1                                              | 24.2 | -20.3      | 24.9 | -21.6      | 25.6 | -23.1      | 26.5 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual                                                            | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |      |
|------|-------------------------------------------------------------------|-------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|
|      |                                                                   | (d)                                                               | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg                                                             | 14.7  | 3.2   | 4.6   | 10.0  | 14.0  | 18.8  | 24.6  | 26.3  | 25.5  | 21.7  | 14.8  | 8.8   | 3.4  | (6)  |      |
| (7)  |                                                                   | DBStd                                                             | 9.68  | 5.81  | 6.53  | 5.66  | 5.28  | 5.03  | 3.13  | 2.83  | 3.08  | 3.82  | 5.30  | 5.49  | 5.65 | (7)  |      |
| (8)  |                                                                   | HDD10.0                                                           | 802   | 221   | 170   | 70    | 22    | 3     | 0     | 0     | 0     | 18    | 85    | 212   |      | (8)  |      |
| (9)  |                                                                   | HDD18.3                                                           | 2218  | 470   | 384   | 262   | 147   | 56    | 1     | 1     | 13    | 133   | 287   | 463   |      | (9)  |      |
| (10) |                                                                   | CDD10.0                                                           | 2513  | 9     | 20    | 70    | 141   | 276   | 438   | 504   | 480   | 351   | 166   | 49    | 8    | (10) |      |
| (11) |                                                                   | CDD18.3                                                           | 887   | 0     | 1     | 3     | 17    | 71    | 188   | 247   | 222   | 114   | 23    | 1     | 0    | (11) |      |
| (12) |                                                                   | CDH23.3                                                           | 10426 | 3     | 22    | 97    | 312   | 884   | 2181  | 2986  | 2411  | 1191  | 296   | 43    | 1    | (12) |      |
| (13) |                                                                   | CDH26.7                                                           | 5154  | 0     | 6     | 23    | 113   | 386   | 1099  | 1639  | 1268  | 519   | 96    | 6     | 0    | (13) |      |
| (14) | Wind                                                              | WSAvg                                                             | 5.5   | 5.2   | 5.5   | 6.0   | 6.7   | 5.8   | 6.0   | 4.9   | 4.6   | 5.1   | 5.3   | 5.4   | 5.1  | (14) |      |
| (15) | Precipitation                                                     | PrecAvg                                                           | 588   | 21    | 16    | 34    | 56    | 86    | 74    | 68    | 78    | 44    | 56    | 25    | 28   | (15) |      |
| (16) |                                                                   | PrecMax                                                           | 863   | 52    | 59    | 148   | 289   | 156   | 241   | 181   | 140   | 109   | 204   | 93    | 107  | (16) |      |
| (17) |                                                                   | PrecMin                                                           | 424   | 0     | 0     | 0     | 0     | 8     | 1     | 9     | 7     | 0     | 6     | 0     | 3    | (17) |      |
| (18) |                                                                   | PrecStd                                                           | 99    | 16    | 16    | 34    | 61    | 37    | 54    | 40    | 35    | 32    | 49    | 27    | 25   | (18) |      |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%                                                              | DB    | 22.6  | 26.2  | 29.0  | 32.7  | 35.8  | 39.2  | 39.8  | 38.2  | 35.8  | 32.6  | 27.4  | 21.4 | (19) |      |
| (20) |                                                                   |                                                                   | MCWB  | 9.4   | 10.3  | 12.8  | 13.8  | 15.4  | 18.7  | 18.6  | 19.7  | 18.4  | 17.0  | 11.8  | 8.8  | (20) |      |
| (21) |                                                                   | 2%                                                                | DB    | 19.9  | 22.2  | 26.2  | 30.1  | 33.0  | 36.4  | 37.2  | 37.0  | 33.9  | 29.1  | 24.0  | 18.0 | (21) |      |
| (22) |                                                                   |                                                                   | MCWB  | 8.6   | 9.4   | 11.9  | 13.1  | 16.1  | 18.9  | 19.2  | 19.7  | 18.6  | 16.5  | 11.4  | 8.1  | (22) |      |
| (23) |                                                                   | 5%                                                                | DB    | 16.1  | 18.9  | 22.8  | 27.3  | 31.2  | 34.6  | 35.8  | 35.1  | 32.2  | 27.0  | 21.1  | 15.4 | (23) |      |
| (24) |                                                                   |                                                                   | MCWB  | 6.8   | 8.2   | 10.8  | 12.9  | 16.2  | 19.3  | 19.6  | 19.9  | 18.6  | 15.2  | 10.5  | 7.0  | (24) |      |
| (25) |                                                                   | 10%                                                               | DB    | 12.7  | 15.4  | 20.1  | 23.9  | 28.4  | 32.4  | 34.1  | 33.0  | 30.0  | 23.9  | 17.8  | 12.5 | (25) |      |
| (26) |                                                                   |                                                                   | MCWB  | 5.0   | 6.6   | 9.8   | 12.4  | 16.2  | 19.3  | 19.9  | 19.6  | 18.1  | 14.5  | 9.3   | 5.7  | (26) |      |
| (27) |                                                                   | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%  | WB    | 10.5  | 12.5  | 15.9  | 17.8  | 21.7  | 23.9  | 23.4  | 23.7  | 21.9  | 20.6  | 15.1 | 11.7 | (27) |
| (28) |                                                                   |                                                                   |       | MCDWB | 21.0  | 22.1  | 22.0  | 23.9  | 27.9  | 32.7  | 30.7  | 32.4  | 29.3  | 25.4  | 19.8 | 16.6 | (28) |
| (29) | 2%                                                                |                                                                   | WB    | 9.0   | 10.2  | 14.2  | 16.3  | 20.2  | 22.3  | 22.4  | 22.5  | 20.9  | 18.9  | 13.0  | 9.2  | (29) |      |
| (30) |                                                                   |                                                                   | MCDWB | 18.0  | 19.9  | 20.8  | 22.6  | 25.9  | 30.4  | 30.4  | 30.6  | 27.6  | 24.6  | 19.0  | 16.4 | (30) |      |
| (31) | 5%                                                                |                                                                   | WB    | 7.3   | 8.8   | 12.8  | 14.9  | 19.0  | 21.5  | 21.6  | 21.7  | 20.2  | 17.6  | 11.7  | 7.6  | (31) |      |
| (32) |                                                                   |                                                                   | MCDWB | 15.6  | 18.2  | 19.2  | 21.4  | 25.7  | 29.5  | 30.1  | 29.4  | 27.1  | 22.7  | 19.0  | 14.3 | (32) |      |
| (33) | 10%                                                               |                                                                   | WB    | 5.5   | 7.3   | 11.2  | 13.6  | 17.9  | 20.7  | 20.8  | 21.0  | 19.6  | 16.1  | 10.3  | 6.1  | (33) |      |
| (34) |                                                                   |                                                                   | MCDWB | 12.0  | 15.3  | 18.5  | 21.2  | 24.5  | 28.6  | 29.4  | 28.5  | 26.5  | 21.6  | 17.5  | 12.2 | (34) |      |
| (35) | Mean Daily Temperature Range                                      |                                                                   | 5% DB | MDBR  | 13.9  | 14.6  | 15.1  | 15.0  | 13.8  | 13.8  | 13.7  | 13.5  | 13.5  | 14.0  | 14.4 | 13.0 | (35) |
| (36) |                                                                   |                                                                   |       | MCDBR | 18.5  | 19.5  | 18.6  | 19.2  | 17.2  | 16.5  | 15.8  | 15.9  | 16.0  | 16.9  | 17.9 | 17.3 | (36) |
| (37) |                                                                   | MCWBR                                                             |       | 9.8   | 9.4   | 8.3   | 7.9   | 6.5   | 5.1   | 4.0   | 4.0   | 4.7   | 6.9   | 8.4   | 9.3  | (37) |      |
| (38) |                                                                   | 5% WB                                                             | MCDWB | 17.7  | 18.5  | 15.8  | 14.4  | 14.1  | 13.9  | 13.6  | 13.3  | 12.5  | 13.1  | 15.4  | 16.2 | (38) |      |
| (39) |                                                                   |                                                                   | MCWBR | 9.6   | 9.3   | 7.9   | 7.4   | 6.7   | 5.7   | 4.5   | 4.3   | 4.5   | 6.4   | 8.0   | 9.4  | (39) |      |
|      |                                                                   |                                                                   |       |       |       |       |       |       |       |       |       |       |       |       |      |      |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub                                                              | 0.257 | 0.264 | 0.294 | 0.326 | 0.342 | 0.378 | 0.381 | 0.371 | 0.339 | 0.303 | 0.276 | 0.257 | (40) |      |      |
| (41) |                                                                   | taud                                                              | 2.592 | 2.587 | 2.502 | 2.425 | 2.444 | 2.382 | 2.412 | 2.404 | 2.475 | 2.564 | 2.567 | 2.596 | (41) |      |      |
| (42) |                                                                   | Ebn at Noon                                                       | 967   | 995   | 981   | 958   | 940   | 901   | 896   | 901   | 919   | 934   | 933   | 945   | (42) |      |      |
| (43) |                                                                   | Edh at Noon                                                       | 74    | 84    | 100   | 114   | 114   | 121   | 117   | 115   | 102   | 85    | 75    | 69    | (43) |      |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg                                                            | 3.05  | 3.83  | 5.04  | 6.24  | 6.91  | 7.39  | 7.20  | 6.43  | 5.49  | 4.14  | 3.22  | 2.63  | (44) |      |      |
| (45) |                                                                   | RadStd                                                            | 0.27  | 0.37  | 0.36  | 0.39  | 0.57  | 0.49  | 0.32  | 0.34  | 0.27  | 0.49  | 0.33  | 0.19  | (45) |      |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (21 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page